

Unit III: Work, Energy, and Power

Work done, kinetic and potential energy, work-energy theorem, conservation of energy, power, and efficiency

Learning Goals

- Understand work as energy transferred by a force causing displacement.
- Calculate work done by constant forces using force, displacement, and angle.
- Distinguish between kinetic energy and potential energy.
- Apply the work-energy theorem to solve motion problems.
- Use conservation of mechanical energy in suitable situations.
- Understand power as the rate of doing work or transferring energy.
- Calculate efficiency and interpret energy losses in real systems.

1. Introduction

The concepts of work, energy, and power provide a powerful way to analyze physical systems. Instead of directly using forces and acceleration at every instant, energy methods often allow us to relate initial and final states more simply.

Energy is the capacity to do work. Work is one way energy is transferred. Power tells how quickly work is done or how quickly energy is transferred.

Work transfers energy

Power measures rate of energy transfer

2. Work Done by a Force

Work is done when a force causes displacement. If there is no displacement, no mechanical work is done by that force, even if the force is large.

$$W = F s \cos(\theta)$$

Symbol	Meaning
W	Work done, measured in joules (J).
F	Magnitude of force, measured in newtons (N).
s	Magnitude of displacement, measured in metres (m).
theta	Angle between force and displacement.

- Work is maximum when force and displacement are in the same direction.

- Work is zero when force is perpendicular to displacement.
- Work is negative when force has a component opposite to displacement.

$$1 \text{ joule} = 1 \text{ newton metre} = 1 \text{ N m}$$

Positive, Negative, and Zero Work

Type of work	Example
Positive work	A force pushes a box in the direction of motion.
Negative work	Friction acts opposite to sliding motion.
Zero work	Centripetal force in uniform circular motion is perpendicular to displacement.

A person holding a heavy bag at rest may feel tired, but in the physics sense no mechanical work is done on the bag if it has no displacement.

3. Work Done by Variable Force

When force changes with position, work is found by adding very small amounts of work over the displacement. Graphically, work is the area under the force-displacement graph.

For variable force: $W = \text{area under } F\text{-}s \text{ graph}$

Calculus form: $W = \int F \, ds$

- For a constant force, the $F\text{-}s$ graph is a rectangle, so $W = Fs$.
- For a spring force, force changes with extension, so work depends on the area under the graph.

4. Kinetic Energy

Kinetic energy is the energy possessed by a body due to its motion. A faster object has more kinetic energy. A heavier object moving at the same speed also has more kinetic energy.

Kinetic energy: $K = \frac{1}{2} m v^2$

- Kinetic energy is a scalar quantity.
- Kinetic energy is always zero or positive.
- Doubling speed makes kinetic energy four times larger.
- The SI unit of kinetic energy is the joule.

5. Potential Energy

Potential energy is stored energy due to position, configuration, or condition. The most common forms in elementary mechanics are gravitational potential energy and elastic potential energy.

Gravitational Potential Energy

Near the Earth's surface, gravitational potential energy depends on mass, gravitational acceleration, and height relative to a chosen reference level.

Gravitational potential energy: $U = mgh$

- The reference level for h can be chosen conveniently.
- Only changes in potential energy usually matter in physics problems.
- Raising an object increases its gravitational potential energy.

Elastic Potential Energy

Elastic potential energy is stored in a stretched or compressed spring. For an ideal spring obeying Hooke's law, the stored energy depends on the spring constant and extension.

Spring force: $F = kx$

Elastic potential energy: $U = (1/2)kx^2$

- k is the spring constant, measured in N/m.
- x is the extension or compression from natural length.
- A stiffer spring has larger k and stores more energy for the same extension.

6. Work-Energy Theorem

The work-energy theorem states that the net work done on a body is equal to the change in its kinetic energy. It connects force-based analysis with energy-based analysis.

$$W_{\text{net}} = \Delta K = K_f - K_i$$

$$W_{\text{net}} = (1/2)m v_f^2 - (1/2)m v_i^2$$

If net work is positive, kinetic energy increases and the object speeds up. If net work is negative, kinetic energy decreases and the object slows down. If net work is zero, kinetic energy remains constant.

- The theorem applies to the net work done by all forces.
- It can be used even when acceleration is not constant, as long as work can be determined.
- It is especially useful when forces vary with position.

7. Conservative and Non-Conservative Forces

A conservative force is a force for which work done depends only on initial and final positions, not on the path taken. Gravitational force and ideal spring force are common conservative forces.

A non-conservative force is a force for which work depends on the path. Friction and air resistance are common non-conservative forces because they dissipate mechanical energy into heat and sound.

Force type	Examples and features
Conservative	Gravity, ideal spring force; path independent; potential energy can be defined.
Non-conservative	Friction, air resistance; path dependent; mechanical energy is dissipated.

8. Conservation of Energy

Energy cannot be created or destroyed, but it can change form. In mechanics, when only conservative forces do work, total mechanical energy remains constant.

Mechanical energy: $E = K + U$

If only conservative forces act: $K_i + U_i = K_f + U_f$

For example, when a ball falls freely, gravitational potential energy decreases while kinetic energy increases. Neglecting air resistance, the total mechanical energy stays constant.

- At the top of a fall, potential energy is high and kinetic energy may be low.
- During the fall, potential energy converts into kinetic energy.
- Just before impact, kinetic energy is maximum and potential energy is lower.

When Mechanical Energy Is Not Conserved

If friction or air resistance acts, mechanical energy decreases, but total energy is still conserved because some mechanical energy becomes thermal energy, sound, or internal energy.

$K_i + U_i + W_{\text{nonconservative}} = K_f + U_f$

- Friction usually does negative work on a sliding object.
- Energy lost from mechanical form appears in other forms.

9. Power

Power is the rate at which work is done or energy is transferred. Two machines may do the same amount of work, but the one that does it faster has greater power.

Average power: $P = W / t$

Power in terms of force and velocity: $P = F v \cos(\theta)$

Quantity	Unit
Power	watt (W).
Work or energy	joule (J).
Time	second (s).
1 watt	1 joule per second.

- Power is a scalar quantity.
- A kilowatt is 1000 watts.
- The commercial unit of electrical energy is the kilowatt-hour.

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

10. Efficiency

Efficiency measures how much useful output energy or power is obtained from the input energy or power. Real machines are never perfectly efficient because some energy is lost, often as heat, sound, vibration, or deformation.

$$\text{Efficiency} = \text{useful output energy} / \text{input energy}$$

$$\text{Percentage efficiency} = (\text{useful output} / \text{input}) \times 100\%$$

- Efficiency is usually less than 100 percent in real devices.
- Reducing friction and heat loss can improve efficiency.
- A more efficient machine wastes less input energy.

For example, if a motor receives 500 J of electrical energy and gives 350 J of useful mechanical energy, its efficiency is $350/500 \times 100\% = 70\%$.

11. Worked Examples

Example 1: Work Done

A force of 20 N moves a box 5 m in the direction of the force. Work done $W = Fs \cos(0) = 20 \times 5 = 100 \text{ J}$.

Example 2: Kinetic Energy

A 2 kg object moves with speed 3 m/s. Kinetic energy $K = (1/2)mv^2 = (1/2)(2)(3^2) = 9 \text{ J}$.

Example 3: Potential Energy

A 5 kg object is lifted to a height of 4 m. Taking $g = 9.8 \text{ m/s}^2$, gravitational potential energy $U = mgh = 5 \times 9.8 \times 4 = 196 \text{ J}$.

Example 4: Work-Energy Theorem

A body of mass 4 kg speeds up from 2 m/s to 6 m/s. Net work = change in kinetic energy = $(1/2)(4)(36) - (1/2)(4)(4) = 72 - 8 = 64 \text{ J}$.

Example 5: Power

If 600 J of work is done in 10 s, average power $P = W/t = 600/10 = 60 \text{ W}$.

12. Common Mistakes

- Forgetting the angle in $W = Fs \cos(\theta)$.
- Assuming work is done whenever force is applied, even when displacement is zero.
- Confusing mass with weight.
- Using conservation of mechanical energy when friction or air resistance is significant without accounting for energy loss.
- Thinking potential energy has an absolute value independent of reference level.
- Forgetting that kinetic energy depends on v squared, not v .
- Confusing power with energy; power is energy transferred per unit time.
- Writing efficiency greater than 100 percent for a real machine.

Quick Revision Sheet

- Work is done when force produces displacement.
- $W = Fs \cos(\theta)$ for a constant force.
- Work is positive, negative, or zero depending on the angle between force and displacement.
- Kinetic energy is $K = (1/2)mv^2$.
- Gravitational potential energy near Earth is $U = mgh$.
- Elastic potential energy is $U = (1/2)kx^2$.
- Net work equals change in kinetic energy.
- Mechanical energy is $K + U$.
- Mechanical energy is conserved when only conservative forces do work.
- Friction and air resistance are non-conservative forces.
- Power is the rate of doing work: $P = W/t$.
- Efficiency compares useful output with total input.

Practice Questions

- Define work done by a force. State its SI unit.
- A force of 50 N moves an object 3 m in the direction of force. Find work done.
- A force acts perpendicular to displacement. What is the work done?
- Calculate the kinetic energy of a 10 kg body moving at 4 m/s.
- Find the gravitational potential energy of a 2 kg body at height 5 m. Take $g = 9.8 \text{ m/s}^2$.
- State and explain the work-energy theorem.

- A 3 kg object speeds up from 2 m/s to 8 m/s. Find the net work done.
- What is meant by a conservative force? Give two examples.
- Why is friction called a non-conservative force?
- State the principle of conservation of mechanical energy.
- A machine does 1200 J of work in 20 s. Find its power.
- Convert 1 kWh into joules.
- A motor has input energy 1000 J and useful output energy 750 J. Find efficiency.
- Why can no real machine be 100 percent efficient?
- Explain the energy transformation when a ball falls freely from a height.